

Name of Student: Vyankatesh Suresh Rohokale	PRN: 26070122518
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 10	
TITLE : Abstract Classes and Methods	

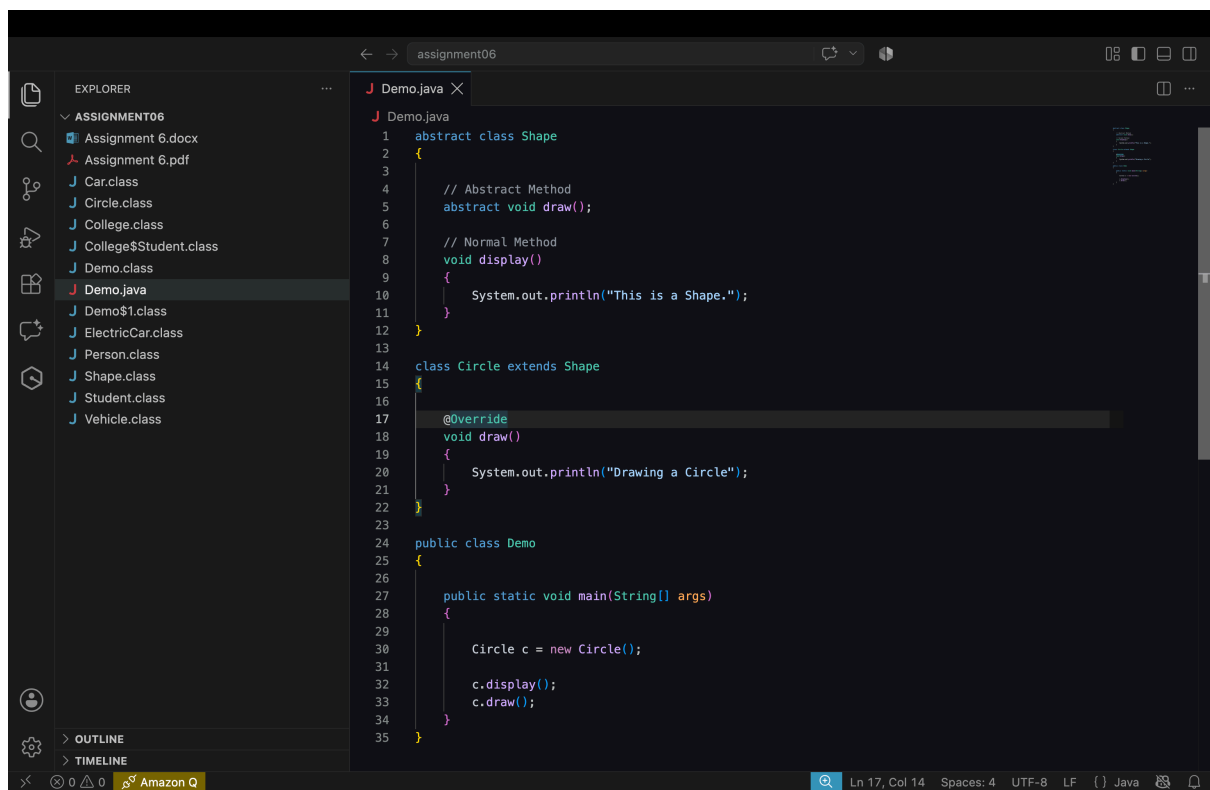
Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

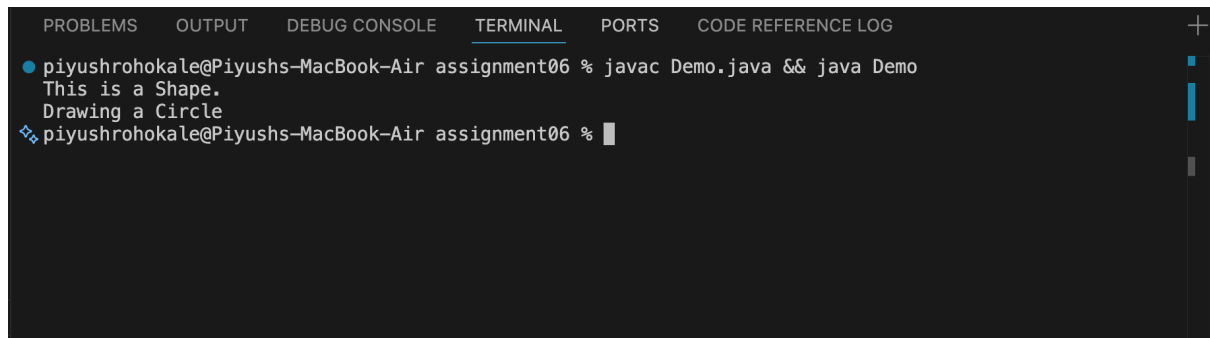
To implement abstraction using abstract classes and abstract methods.

Programme



```
1  abstract class Shape
2  {
3
4      // Abstract Method
5      abstract void draw();
6
7      // Normal Method
8      void display()
9      {
10         System.out.println("This is a Shape.");
11     }
12 }
13
14 class Circle extends Shape
15 {
16
17     @Override
18     void draw()
19     {
20         System.out.println("Drawing a Circle");
21     }
22 }
23
24 public class Demo
25 {
26
27     public static void main(String[] args)
28     {
29
30         Circle c = new Circle();
31
32         c.display();
33         c.draw();
34     }
35 }
```

Output

A screenshot of an IDE's terminal window. The terminal has a dark background with light-colored text. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active and underlined), 'PORTS', and 'CODE REFERENCE LOG'. The terminal shows the command 'javac Demo.java && java Demo' being executed. The output of the program is displayed on the next two lines: 'This is a Shape.' and 'Drawing a Circle'. The prompt 'piyushrohokale@Piyushs-MacBook-Air assignment06 %' is visible at the end of the first line and before the second line.

```
piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
This is a Shape.
Drawing a Circle
piyushrohokale@Piyushs-MacBook-Air assignment06 %
```

Conclusion

The program successfully demonstrates the use of an abstract class and an abstract method in Java. The abstract class provides a common blueprint , while the child class implements the abstract method according to its requirements. This approach supports abstraction , improves code reusability , and allows a common interface for related classes.